

How Far Do Surface Cues Go?

Baselines for SemEval-2027 Task 2 (DiCo-NLI)

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Abstract

SemEval-2027 Task 2 (DiCo-NLI) hands a system two short phrases in a fixed order and asks which of four relations holds between them: equivalence, entailment in either direction, or none. It also checks whether the system’s answer on a pair agrees with its answer when the pair is turned around. We wanted to know how much of this a model can get right without reading for meaning at all. On the English track, surface form already gives direction away. In 78% of forward-entailment training instances the premise is the longer phrase, and in 45% every word of the hypothesis already appears in the premise. A logistic regression on 17 cues of this kind reaches weighted F1 0.531 on the development set, and it is the most self-consistent of our classical baselines, with SOFTCONS 0.794. A fine-tuned DeBERTa-v3-base does far better, 0.807 ± 0.015 weighted F1 over three seeds with SOFTCONS 0.877 and HARDCONS 0.810, and its gain comes almost entirely from the instances where length says nothing and from the line between equivalence and entailment. What it still gets wrong are prepositions, numerals and single substituted words. We also note that a system which answers EQUIVALENCE to everything gets a perfect SOFTCONS, so none of the three official scores means much on its own.

1 Introduction and Task Definition

The task as a black box. DiCo-NLI (DiCo-NLI organizers, 2026) is four-way classification over ordered pairs of short phrases. The input is a premise p and a hypothesis h , about three words each. The output is one label from $\mathcal{L}$: EQUIVALENCE (the phrases mean the same thing), FORWARD (p entails h), BACKWARD (h entails p) or NEGATIVE (none of these; Table 1). Take “A young boy” and “A boy”. The first entails the second, so the ordered pair is FORWARD, and the same two phrases in the other order are BACKWARD. The organisers

Input	ordered phrase pair (p, h) , English, about three words each
Output	one label per ordered pair
Label space	EQUIVALENCE, FORWARD, BACKWARD, NEGATIVE
Formulation	four-way classification with a cross-instance constraint between (p, h) and (h, p)
Official scores	weighted F1, SOFTCONS, HARDCONS
Our data	track 1 (English): train 3,042, dev 660 instances; dev serves as our test set

Table 1: DiCo-NLI track 1 at a glance.

build that reversal into the data. Every reversible source pair occurs twice, once in each order, and a fixed operator ties the two gold labels together: EQUIVALENCE stays EQUIVALENCE, FORWARD and BACKWARD swap. Three scores are reported side by side. Weighted F1 is the usual per-instance score. SOFTCONS asks whether a system’s two answers on a pair obey the reversal operator, and HARDCONS whether both of them are correct; Section 7 gives the formulas. A system sees only the two strings and their language codes. The instance and pair identifiers are bookkeeping, and as we explain below they have to stay out of the model.

The task as a white box. Underneath, the task is about meaning containment. In natural logic, adding a restrictive modifier narrows a phrase, and the narrower phrase entails the broader one (MacCartney and Manning, 2009); “on a red couch” entails “on a couch”. So the cues that ought to matter are which phrase carries the extra material, whether the two heads match, and what quantifiers, numerals and prepositions do to the scope of a phrase. The same structure also opens a shortcut. If the more specific phrase is usually the longer one, then length and word containment predict direction without any semantics at all, and Section 4 shows that in this data they do.

Three further things shaped how we set the exper-

iments up. The instance identifier leaks the label, because only reversible pairs carry a `__flipped` twin, so no model input may include it. About a fifth of the development phrases also occur somewhere in training. And the release does not say whether a pair came from an image caption or a news headline, so we cannot stratify by source.

Why the task is hard. There is very little to go on. Three words, and no sentence around them. Telling equivalence from entailment often needs lexical knowledge (is “a chocolate bar” the same as “a candy bar”, or a kind of one?). Negatives are frequently a single swapped word in an otherwise identical frame (“off Mexico” against “on Mexico”). And consistency is a property of two predictions that a classifier makes separately, with no idea that they belong together.

Question and findings. Our question is whether systems reach directional consistency by representing the relation between the two phrases, or by picking up surface asymmetries, and how much of a fine-tuned encoder’s three scores a cue-only model recovers. We fixed three hypotheses before running anything. H1: a cue-only model reaches a large share of the encoder’s SOFTCONS, because each of its cues is either symmetric or flips sign exactly under reversal. H2: the encoder’s advantage concentrates where the cues are uninformative, including the boundary between equivalence and entailment. H3: accuracy and consistency trade off in models trained on single ordered instances. The first two held. The third held inside our classical baselines and failed for the encoder, which changes what Assignment 2 should aim at.

2 Related Work

The pilot study. The task grows out of [Apao-laza Larraya et al. \(2026\)](#), who defined the two consistency measures (they call them soft and hard causal coherence) and ran encoder, decoder and encoder-decoder transformers on PhrasIS phrase pairs, each tuned with up to 150 Optuna trials on an 80/20 split of the training data. Encoders led throughout. DeBERTa-v3-base reached weighted F1 0.75, soft coherence 0.84 and hard coherence 0.79 on the combined positives track; decoder-only models trailed on both accuracy and coherence, and every family lost ground on pairs it had not seen. Their manual error analysis points at prepositional phrases and at near-synonym or taxonomic rela-

tions. We cite these numbers more than once, but their track has no negatives and a different split, so for us they are a reference point and not a target.

Direction and the reversal curse. The framing comes from [Berglund et al. \(2024\)](#): an autoregressive language model fine-tuned on “A is B” cannot answer “B is A”. [Lv et al. \(2024\)](#) trace this to the next-token objective and its causal attention, and repair it at training time. DiCo-NLI is a gentler setting, since both phrases are on the table, yet the pilot’s decoder results suggest the asymmetry survives. Direction is fragile elsewhere too. [Yanaka et al. \(2019\)](#) found that NLI models handle upward monotonicity inferences far better than downward ones, and at phrase level that is exactly the forward against backward distinction.

Consistency as a constraint. [Minervini and Riedel \(2018\)](#) and [Li et al. \(2019\)](#) turned logical relations among groups of NLI examples, such as the symmetry of contradiction and transitivity, into training penalties, and improved accuracy and consistency together. [Elazar et al. \(2021\)](#) define consistency as invariance under meaning-preserving rewording and measure it on ParaRel. Reversal belongs to the same family of constraints, except that the label has to change in a prescribed way rather than stay put.

Data and models. The phrases are from PhrasIS ([Lopez-Gazpio et al., 2024](#)), which annotates phrase pairs taken from the chunk alignments of interpretable semantic textual similarity ([Agirre et al., 2016](#)). That origin is why the items are noun phrases, verb chains and prepositional phrases rather than sentences. Our encoder is DeBERTa-v3-base ([He et al., 2023](#)), the strongest base-size model in the pilot.

3 Dataset

We work with the track 1 (English) training and development files of the official release at commit 588968e (8 September 2026). A reference row holds an instance identifier, a source-pair identifier, the two phrases with their language codes, the identifier of the reversed twin, and the gold label. Table 2 has the split statistics and the quality checks we ran.

Splits. The official test set will not appear until 10 January 2027, after the course ends, so the development set is our test set and we treat it that

	Train	Dev
Instances	3,042	660
Source pairs	1,760	383
reversible (two instances)	1,282	277
negative (one instance)	478	106
Labels E/F/B/N (%)	26/29/29/16	26/29/29/16
Words per phrase, mean (max)	3.0 (15)	2.9 (10)
Missing or empty texts	0	0
Duplicate pairs, groups (rows)	43 (51)	4 (4)
with conflicting labels	1	0
One-word phrases	563	135
Phrases of 8+ words	27	7
Phrases starting with a clitic	56	21
Language pair	en, en	en, en

Table 2: Track 1 splits and quality checks. E/F/B/N: EQUIVALENCE, FORWARD, BACKWARD, NEGATIVE. Duplicate rows are extra copies of an ordered pair; phrase counts are occurrences over both sides of every instance; clitic phrases include those starting with punctuation.

way: nothing is selected on it. Model selection uses slices of the training data, split by source pair so that both directions of a pair land on the same side. Train and dev share no source pair. They do share text, though. 7 unordered phrase pairs and 155 of the 712 distinct development phrases (22%) also occur in training. We report this rather than hide it, and we come back to it when we read the scores.

Quality. The files are complete, with no missing or empty text. The one genuine defect we found is small. “South Africans” paired with “South Africa” appears twice in training, once as FORWARD and once as NEGATIVE, and both cannot be right. Beyond that, one-word phrases are common (563 occurrences in train), and some phrases are visibly fragments of the chunking that produced them, “s war shrine visit” for instance. Whether a pair came from an image caption or a news headline is not recorded in the train and dev files. Only the 461-pair trial sample carries that field (244 caption and 217 headline pairs), so we cannot stratify by source.

4 Exploratory Data Analysis

We ran the analysis on the training split only and organised it around six feature groups, each chosen to answer one question about the task (Table 3). We left out lexical-richness measures such as TTR, MTLT and HD-D on purpose: on three-word phrases they are undefined, or they jump around with a single word. The seven figures of

Group	Question
Length	Does the longer phrase predict the direction?
Readability	Do word and syllable lengths differ by relation?
Lexical	Do overlap and containment separate direction, equivalence and negatives?
Semantic	Does embedding similarity separate the relations, and can it see direction?
Linguistic	Do heads, POS patterns, adjectives and prepositions vary by relation?
Task cues	Are quantifiers, negation, numerals and named entities frequent enough to matter?

Table 3: The six EDA feature groups and the question each answers.

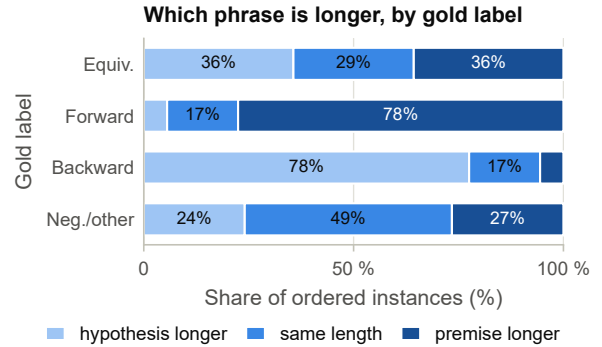


Figure 1: Which phrase is longer, by gold label (train).

the analysis are collected, without captions, in the separate EDA PDF. Two of them appear here.

Length. *Observation.* The premise is the longer phrase in 78% of FORWARD instances and the hypothesis in 78% of BACKWARD instances. EQUIVALENCE splits almost evenly (36% premise longer, 29% equal, 36% hypothesis longer), and about half of the NEGATIVE instances (49%) have the same length on both sides (Figure 1). *Interpretation.* Direction leaks through form. The entailing phrase is usually the more specific one, and specificity in this data is mostly added words (“in Russia” against “in Russian city of Kazan”), so a model can learn direction without touching meaning. *Implication.* Length goes into the cue baseline. Every model also gets scored separately on the instances where length points the wrong way or nowhere (Section 9), because those are the ones that tell us whether anything beyond the shortcut was learned.

Lexical overlap and containment. *Observation.* Mean word-set Jaccard overlap is 0.41 for directional pairs, against 0.24 for EQUIVALENCE and 0.27 for NEGATIVE. Containment is one-sided: every hypothesis word occurs in the premise in 45%

of FORWARD instances and only 2% of BACKWARD instances, and the mirror holds for the other direction. The last word is shared in 58% of directional instances (Figure 2). *Interpretation.* A directional pair is typically one phrase with words added or removed around a head that stays put. Equivalences look different. They are paraphrases, often with little literal overlap: “Woman and dog” and “an adult female accompanied by a dog” share one word. *Implication.* Containment in both orientations is a strong direction cue when it fires, but it fires on fewer than half of the directional pairs. Overlap tells a model that two phrases are related; it does not tell it that they are equivalent.

Semantic similarity. *Observation.* Cosine similarity of MiniLM sentence embeddings (Reimers and Gurevych, 2019) has a median of 0.76 for EQUIVALENCE, 0.77 for both directions and 0.64 for NEGATIVE. 52% of directional and 50% of EQUIVALENCE instances sit at or above the EQUIVALENCE median, against 27% of negatives. Cosine is symmetric, so a pair and its reversal get exactly the same value. *Interpretation.* Similarity separates negatives from related pairs, moderately, and does nothing for the line between EQUIVALENCE and entailment. It carries no information about direction, by construction. *Implication.* We kept cosine out of the cue set. A symmetric representation cannot recover direction, and the equivalence boundary needs something that reads the ordered pair, which is what we expected the encoder to supply.

Linguistic and task-specific cues. *Observation.* The two heads share a lemma in 52% of directional instances against 30% of EQUIVALENCE. The part-of-speech pattern is identical on both sides in 31% of NEGATIVE instances and only 7% of directional ones. The premise of a FORWARD pair carries 0.33 more adjectives, on average, than its hypothesis. Numerals occur in 18% of directional instances and named entities in 47%; quantifiers (3%) and negation (under 1%) are rare. *Interpretation.* A negative is often one substituted word inside an unchanged frame (“Two young children” against “Two small children” is labelled NEGATIVE), and adjectives are the material that narrows a phrase. *Implication.* Comparing heads and modifiers should matter more than any global similarity. Quantifiers and negation are too rare to train on, so we keep them as error-analysis slices instead.

Group	Cues
Length (7)	words and characters per side, their differences, log word ratio
Overlap (2)	shared words, Jaccard
Containment (4)	share of each side’s words in the other; each side contained as a contiguous span
Position (2)	same first word, same last word
Numerals (2)	a digit on each side

Table 4: The 17 surface cues, grouped.

Readability. *Observation.* Syllables per word run from 1.53 to 1.60 across labels, characters per word from 4.5 to 4.9, and median Flesch reading ease from 77.9 to 82.4. *Interpretation.* The phrases are uniformly short caption and headline language. Flesch assumes sentences, and on fragments it only tracks syllable density. *Implication.* Readability does not separate the labels and stays out of the feature set. It stays in the report so that nobody has to rerun the check.

5 Feature Engineering

The classical baselines see two feature sets, and both are computed from the two phrase strings alone. A unit test checks that no identifier reaches the feature matrix, since the instance identifier would otherwise give the label away.

Surface cues. Seventeen numeric cues turn the EDA findings into features (Table 4). We chose them so that each behaves predictably when the pair is reversed: overlap and shared-word counts do not change, length differences change sign, and the two containment cues swap places. That property is what makes the cue model consistent almost by construction, and Section 8 leans on it. Readability and embedding similarity stay out, for the reasons given in Section 4. The cues are standardised on the training data.

Lexical n-grams. TF-IDF over word unigrams and bigrams of the string “*p* [SEP] *h*”, with sub-linear term frequency. The token pattern keeps one-letter words and treats the separator as a token in its own right, so the bigrams that straddle it record which side a word came from. Without that, the bag of words would be almost blind to order.

6 Baseline Models

Every setting below was fixed before the first run. Nothing was tuned on the development set.

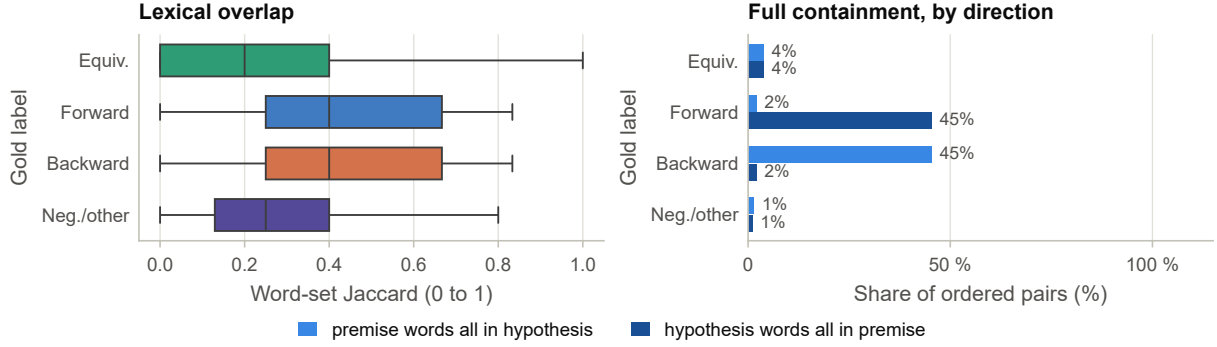


Figure 2: Word overlap (left) and full containment in each direction (right), by gold label (train).

Trivial systems. A majority-label predictor, a uniform random predictor with seed 13, and two constant predictors that answer EQUIVALENCE everywhere and NEGATIVE everywhere. The majority system has a quirk worth recording: FORWARD and BACKWARD tie at 881 training instances each, the tie goes to the first label in canonical order, and so it predicts FORWARD for everything. The two constant systems exist to probe the metrics (Section 7).

Classical baselines. Logistic regression on the cues, on the n-grams, and on both together, plus gradient-boosted trees on the cues, all in scikit-learn (Pedregosa et al., 2011) with balanced class weights. The regressions use $C = 1$; the trees run 300 iterations at learning rate 0.05; everything else is a library default.

Pretrained baseline. DeBERTa-v3-base (He et al., 2023) with a classification head, fully fine-tuned with Hugging Face Transformers (Wolf et al., 2020) in a training loop we wrote ourselves so that the seed controls everything. The pair is tokenised as one sequence of at most 32 tokens. Training runs 5 epochs of AdamW at learning rate 2×10^{-5} , batch size 16, weight decay 0.01, a linear schedule with 6% warm-up and gradient clipping at 1.0, and we keep the epoch with the best weighted F1 on a 20% slice of the training data, again split by source pair. Three seeds (13, 42, 1234), each drawing its own slice. One detail tripped us up: under Transformers 5 the checkpoint loads in float16, and the first GPU epoch trained with a NaN loss from its first step until we forced float32. After that, each run trained in about 140 seconds on one NVIDIA T4, and a second execution reproduced every development prediction exactly.

7 Evaluation Setup

Weighted F1. With n_c gold instances of label c among N , and F_c the F1 of label c ,

$$\text{wF1} = \sum_{c \in \mathcal{L}} \frac{n_c}{N} F_c, \quad F_c = \frac{2P_c R_c}{P_c + R_c}. \quad (1)$$

Each label counts in proportion to how often it occurs. So NEGATIVE, 16% of the data and the hardest class for every one of our models, weighs the least. That is why we print macro F1, which averages the F_c with equal weights, next to the official score: a weak minority class should stay visible. Micro F1 would equal accuracy here, since every instance has exactly one label.

Consistency. Let Rev map EQUIVALENCE to EQUIVALENCE, FORWARD to BACKWARD and BACKWARD to FORWARD; NEGATIVE has no reverse. Let $\mathcal{P}$ be the set of reversible pairs (x, x^r) in the development set ($|\mathcal{P}| = 277$), f the system, y the gold label and $\mathbb{I}[\cdot]$ the indicator:

$$\text{SOFTCONS} = \frac{1}{|\mathcal{P}|} \sum_{\mathcal{P}} \mathbb{I}[f(x) = \text{Rev}(f(x^r))], \quad (2)$$

$$\text{HARDCONS} = \frac{1}{|\mathcal{P}|} \sum_{\mathcal{P}} \mathbb{I}[f(x) = y \wedge f(x^r) = y^r]. \quad (3)$$

Since the gold labels obey the reversal themselves, a pair that is right in both directions is automatically consistent, so $\text{HARDCONS} \leq \text{SOFTCONS}$ always. Negative pairs have no reversed twin and enter weighted F1 only. A NEGATIVE prediction on a reversible pair counts as incompatible.

What each score rewards. Weighted F1 rewards correct labels on single instances. SOFTCONS rewards agreeing with yourself, right or wrong. HARDCONS rewards being right on whole pairs. The constant systems make the difference concrete. Answer EQUIVALENCE everywhere and you are consistent on every single pair: SOFTCONS 1.000,

weighted F1 0.110, and a HARDCONS of 0.314 that is simply the share of reversible pairs whose gold label is EQUIVALENCE (87 of 277). Answer NEGATIVE everywhere and you get 0.044, 0 and 0. Neither system knows anything. This is why we never quote one of the three scores on its own.

Protocol. Every score comes from the official scorer, run unchanged, the way the competition platform runs it. The classical baselines are also cross-validated on the training data, five folds grouped by source pair, each fold scored by the same scorer. Each run records its seed, the data commit, the code commit and the exact command, and Section A maps every number in this report back to its run folder.

8 Results

Table 5 lists every system on the development set. Two things first. Cross-validation on the training data ranks the classical systems in the same order as the development set does. And every development score lands at or below its cross-validated mean, by 0.1 to 1.2 standard deviations, and by 2.0 for n-grams alone, so the development set is somewhat harder than held-out training folds. Nothing was selected on it, so we take the ranking at face value and the absolute numbers with a little salt.

Surface cues buy consistency (H1). The cue-only model is the most self-consistent classical system. 220 of the 277 reversible pairs get compatible answers, and across the 380 directional instances it confuses FORWARD with BACKWARD only 8 times. Its weights (Table 6) read like a summary of the EDA: hypothesis-in-premise containment pushes towards FORWARD and away from BACKWARD, its mirror does the opposite, and the length differences add a smaller push the same way. All of these swap or change sign when the pair is reversed, so the model’s two decisions on a pair agree with each other almost by construction. Agreeing is not the same as being right, though. 67 of those 220 consistent pairs are wrong in both directions, which is why HARDCONS stops at 0.552.

Inside the classical family, accuracy costs consistency (H3). Add the n-grams to the cues and weighted F1 climbs from 0.531 to 0.596 while SOFTCONS drops from 0.794 to 0.747. Trees on the same cues make the same trade (0.574 and 0.718). The more expressive models find order-specific patterns that help on single instances and

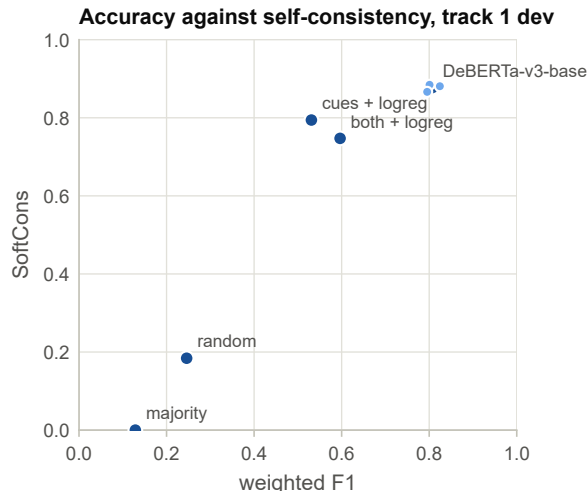


Figure 3: Weighted F1 against SOFTCONS on the development set; small points are the three DeBERTa seeds.

break the agreement across a pair. N-grams on their own see order only through the bigrams that cross the separator, and they score lowest on every metric.

The encoder gains on all three (H3 fails for it). DeBERTa-v3-base adds 0.21 weighted F1 and 0.24 HARDCONS over the best classical system, and 0.08 SOFTCONS over the most consistent one (Figure 3). The gain lands where the cues were weakest. Against the combined classical model, F1 on EQUIVALENCE goes from 0.45 to 0.78 and on NEGATIVE from 0.41 to 0.63, while the two directions move from 0.72 and 0.71 to 0.86 and 0.87. Its consistency is a by-product of correctness: depending on the seed, only 16 to 21 of its consistent pairs are wrong. We make no claim against the pilot’s numbers, which come from another track, another split and a 150-trial search.

9 Error Analysis

How pairs fail. Table 7 sorts each reversible development pair by what a system did with its two directions. The cue model almost never gets one direction right and the other wrong; that happens on a single pair. Its errors are symmetric, either consistent and wrong (67 pairs) or wrong both ways (56). Adding n-grams turns 21 pairs into half-right ones, and that is the order-specific behaviour behind its lower SOFTCONS. DeBERTa leaves 49 to 56 pairs per seed short of fully correct, spread fairly evenly across the three ways of failing.

Where the encoder wins (H2). We split the development instances by what the length cue says

System	Weighted F1	Macro F1	SOFTCONS	HARDCONS	CV weighted F1
Majority (FORWARD everywhere)	0.129	0.112	0.000	0.000	
Random, seed 13	0.246	0.240	0.184	0.054	
EQUIVALENCE everywhere	0.110	0.104	1.000	0.314	
NEGATIVE everywhere	0.044	0.069	0.000	0.000	
Cues, logistic regression	0.531	0.502	0.794	0.552	0.575 ± 0.037
N-grams, logistic regression	0.407	0.399	0.426	0.256	0.429 ± 0.011
Cues and n-grams, logistic regression	0.596	0.573	0.747	0.570	0.617 ± 0.021
Cues, gradient-boosted trees	0.574	0.550	0.718	0.527	0.572 ± 0.024
DeBERTa-v3-base, mean $\pm$ sd of 3 seeds	0.807 $\pm$ 0.015	0.786 $\pm$ 0.017	0.877 $\pm$ 0.010	0.810 $\pm$ 0.013	
<i>Pilot DeBERTa-v3-base, other track[†]</i>	<i>0.75</i>	<i>n/a</i>	<i>0.84</i>	<i>0.79</i>	

Table 5: Track 1 development set, official scorer. CV: cross-validation on the training data, five folds grouped by source pair, mean $\pm$ sd. For DeBERTa the held-out weighted F1 at the selected epoch averages 0.798. [†]Reported by [Apaolaza Larraya et al. \(2026\)](#) on the PhrasIS positives track, without negatives and on another split; a reference point, not comparable.

Cue	E	F	B	N
h words in p	-0.09	+0.96	-1.44	+0.57
p words in h	-0.02	-1.35	+0.97	+0.41
Jaccard overlap	-0.42	+0.84	+0.82	-1.25
Character difference	+0.01	+0.49	-0.49	-0.01
Log word ratio	-0.02	+0.39	-0.38	+0.02

Table 6: The five largest standardised weights of the cue-only logistic regression, per label (E/F/B/N: EQUIVALENCE, FORWARD, BACKWARD, NEGATIVE).

System	Consistent		Inconsistent	
	right	wrong	one right	both wrong
Cues	153	67	1	56
Cues and n-grams	158	49	21	49
DeBERTa, seed 13	228	16	15	18
DeBERTa, seed 42	224	21	12	20
DeBERTa, seed 1234	221	19	17	20

Table 7: Outcomes on the 277 reversible development pairs. A consistent pair is either right in both directions or wrong in both.

about the gold label (Table 8). It agrees with the label on 282 instances, points the wrong way on 10, and says nothing on 368, either because the two phrases have the same length or because the label has no direction. Where it agrees, the cue model is already strong and the encoder adds just 0.02 weighted F1. Where it says nothing, the cue model collapses to 0.31 and the encoder holds at 0.75. That gap of 0.44 is most of the encoder’s advantage. The misleading slice has 10 instances, and we read nothing into it. Slice scores are plain classification scores, because a slice can separate the two directions of a pair, which makes the consistency measures meaningless on it.

Length cue	n	Cues	Cues + n-grams	DeBERTa
Agrees	282	0.907	0.917	0.931
Misleads	10	0.000	0.000	0.800
Uninformative	368	0.313	0.425	0.754

Table 8: Weighted F1 on development slices defined by the length cue; DeBERTa is the mean of three seeds.

What the encoder still gets wrong. Direction is rarely the problem: seed 13 swaps FORWARD and BACKWARD on 3 instances. The errors sit on the EQUIVALENCE and NEGATIVE boundaries instead. 26 directional instances are predicted as EQUIVALENCE, 19 EQUIVALENCE instances as NEGATIVE, and 34 of the 106 negatives are called related. All three seeds miss the same 80 instances, and the cue model misses 50 of those too, so a hard core of the data resists both approaches. We read 20 errors of seed 13 by hand and sorted them into five categories modelled on the pilot’s manual analysis (Table 9). We picked them, four per category, so the counts illustrate the kinds of error and say nothing about how often each occurs. “at a pool” against “in front of a pool” turns on a preposition. “A group of people” against “Six people” turns on cardinality. “on a crowded street” against “on a public street” keeps every word but one, and the gold label is NEGATIVE.

Some errors may be in the gold labels. “two people” and “2” are labelled NEGATIVE. “Saudi Women” to “Saudis” is labelled BACKWARD, though the first phrase is the more specific one. Put these next to the conflicting “South Africans” duplicate in training (Section 3) and it seems likely that some share of the remaining errors are anno-

Category	Premise	Hypothesis	Gold	Predicted
Preposition	at a pool	in front of a pool	BACKWARD	NEGATIVE
Numeral or quantifier	A group of people	Six people	FORWARD	EQUIVALENCE
Near-synonym or taxonomy	a chocolate bar	a candy bar	FORWARD	EQUIVALENCE
One-word negative contrast	on a crowded street	on a public street	NEGATIVE	EQUIVALENCE
Chunking fragment	in destroying	destroy	FORWARD	EQUIVALENCE

Table 9: One example per error category, DeBERTa seed 13 on the development set.

tation disagreements rather than model mistakes. We have not audited the labels systematically. We say so because it limits how much weight the error categories can carry.

10 Discussion

Direction is the easy part. Seventeen surface cues decide FORWARD against BACKWARD almost as reliably as a fine-tuned encoder: 8 confusions against 3, out of 380. What the cues cannot decide is the kind of relation. Does the added material narrow the meaning or paraphrase it? Does a swapped word break the relation or keep it? That is where most of the encoder’s 0.21 gain in weighted F1 comes from (Table 8), and it is also where the encoder’s own errors remain.

SOFTCONS on its own is easy to game. A constant EQUIVALENCE predictor scores a perfect 1.000. The cue model reaches 0.794 while being wrong in both directions on a quarter of the pairs. A system can be consistent by construction, through symmetric or sign-flipping inputs, without any grasp of the relation. We therefore read SOFTCONS only next to HARDCONS and macro F1, and we count the consistent-but-wrong pairs explicitly. We would suggest other participants do the same.

The encoder is consistent because it is correct. The trade between accuracy and consistency that we saw inside the classical family did not carry over. DeBERTa improved all three scores at once, and almost all of its consistent pairs are correct ones. For decoder models, which the pilot found far less coherent, this sharpens the question for Assignment 2: their consistency gap may be a correctness gap on the hard relations, and not a separate defect that needs its own fix.

How far the numbers travel. The development set is a little harder than held-out training folds, a fifth of its phrases also occur in training, and some gold labels look questionable. Differences of a few points between systems should be read with all of

that in mind. The differences we draw conclusions from are 0.2 or more.

11 Next Steps

Assignment 2 moves to decoder language models, where the pilot found the largest consistency gap. We plan to compare open-weight instruction-tuned models used by prompting and fine-tuned with low-rank adapters, and to test three ways of giving a model direction: training on both orders of every pair, a loss term that penalises incompatible predictions on a pair and its reversal in the manner of Li et al. (2019), and prompts that show both orders at once. Every system will be reported with weighted F1, macro F1, SOFTCONS, HARDCONS, the consistent-but-wrong count and the length-cue slices of Table 8, against the baselines in this report. Before we lean on the error categories any further we will audit the conflicting and questionable gold labels. The Spanish and Basque tracks come in if time allows.

Limitations

Everything here is on one track, English, and on the development set, which stands in for the unreleased test set. The classical baselines use one seed and fixed settings; the encoder uses three seeds and a fixed recipe. Nothing was tuned, so these numbers are baselines, not upper bounds. The misleading length-cue slice has 10 instances. The error categories come from 20 hand-picked errors and describe kinds of failure, not their frequency. The part-of-speech and similarity features depend on spaCy’s small English model and on one sentence encoder, both run on phrases with no context around them.

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A Where Every Number Comes From

Each run folder under experiments/ holds a config.json with the seed, the data commit (588968e), the code commit and the exact command, next to the official scorer’s reports. The notebook in notebooks/assignment-1/ that produced each folder:

Notebook	Run folder
01_trivial_baselines	2026-09-29-trivial-baselines-track1
02_eda_track1	2026-09-29-eda-track1
03_ml_baseline_track1	2026-09-29-ml-baseline-track1
04_dl_baseline_track1	2026-09-29-deberta-base-track1
05_error_analysis_track1	2026-10-02-error-analysis-track1